Game design document

Written 2026-10-12, after WP-C3. Revised the same day after your first read. You decided four things then, none of them built yet: the board is seen from straight above, during a wave too (§3.1); walls are single tiles and no longer tetromino pieces (§4.1); the map has rocks (§3.2); and the board draws a route line for every kind of enemy you have met (§5.3, which also says where enemies walk). No name has been chosen for the game yet (§12, question 12).

This file describes the game from the player's side: what is on the screen, what the player does, and what the finished game is meant to be. The other documents answer other questions: `CLAUDE.md` says what the thesis claims and what must never be broken, `ARCHITECTURE.md` how the code is built, `WORKPLAN.md` in what order. Where this file and `CLAUDE.md` disagree, `CLAUDE.md` wins.

Much of the finished game is not decided yet. So every part of this file carries one of four marks:

Mark	Meaning
BUILT	In the game today, and tested.
PLANNED	Decided and in the work plan. Not built.
PROPOSED	A suggestion made in this file, so that there is something concrete to look at. Not decided. Accept it, change it or drop it.
OPEN	Nobody has decided. §12 lists every open question, with who must answer it and by when.

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1. The game in short

A single-player maze tower defense on one square board, seen from straight above. Enemies come out of a spawn in one corner and head for your core in the opposite corner. You have one budget and spend it on two things:

- **walls** (single tiles that you draw onto the board), which bend the enemies' route;
- **towers** (bought from a shop that offers four at random), which shoot them.

The unusual rule: **nothing you build blocks the path.** Every wall and every tower can be chewed through. An enemy takes the cheapest way to the core, and "chew through this" is priced into that choice like any detour. So you never close a route; you make it long and expensive, and you put towers along it.

Waves come one after another, each stronger than the last. Between waves you build, buy and sell. The session ends when ten enemies have reached the core.

Not in today's build yet: the view from above and the single-tile walls. Today the camera is a tilted 3-D one and walls are tetromino pieces dealt from a bag (§3.1, §4.1).

What the thesis adds. The game exists in two versions that differ in one thing only. In the *static* version every wave is the same kind of enemy, just more and tougher. In the *adaptive* version a wave director looks at what you built, picks the kind of wave that presses on that build's weak point, and tells you so in one line. Both versions send waves of the same total strength. The claim to be tested is that the adaptive version keeps a player rebuilding, where the static one lets them find one good maze and repeat it.

Design pillars

1. **You bend the path, you never close it.** A path to the core always exists.
2. **Two levers, one budget.** Towers are damage. Walls are time under fire. Walls alone kill nothing.
3. **Every wave asks a question about your build**, and says the question out loud.
4. **Fair.** Whatever the director sends, it has the same strength as the static wave of that number, and the director never touches your shop, your prices or your money.
5. **Readable.** The routes are drawn (§5.3). A tower shows its range before you pay for the tile. An enemy shows its health. Nothing important is hidden in a number.

2. One wave, from start to finish

```
BUILD PHASE      45 seconds before wave 1, 10 seconds after every wave
  │              draw walls · buy towers and place them · sell what did not work
  │              Start Wave skips the rest of the clock
  ▼
THE WAVE STARTS  the banner says what is coming
  │              enemies leave the spawn one after another
  │              towers fire · enemies walk the route, or chew through what is in the way
  │              you may still place walls and buy towers; you may not sell
  ▼
THE WAVE ENDS    when the last enemy has been killed or has reached the core
                 +20 budget · four new shop offers · the next build phase begins
```

All of this is BUILT, except the banner (PLANNED , WP14).

3. The board

BUILT One map, 38 × 38 tiles, with nothing on it. The spawn is near one corner (tile 3, 3) and the core near the opposite one (tile 34, 34), so with nothing built the enemies walk the diagonal. At wave-1 speed that walk takes a little over a minute. Any tile can be built on, except the spawn's and the core's.

3.1 Seen from straight above

PLANNED (your decisions, 2026-10-12; in the work plan under WP-C4). The board lies flat on the screen and all of it is visible at once, as in the pictures of §5.3 and §9. There is no camera to turn or move, so four controls go away (§10).

The wave is watched in the same view as the build phase. One view to learn is less than two, and a participant has a single sitting to learn the game in. PROPOSED a flyer gets a shadow under it, so that "above the walls" can still be read from above.

BUILT **today:** a tilted 3-D camera that can be turned, moved, zoomed and rolled (§9.1).

3.2 Rocks

PLANNED (your decision, 2026-10-12; in the work plan under WP-C4). An empty board of 1,444 equal tiles gives a new player nothing to start from. So the map gets **rocks**: fixed tiles that belong to the map and not to the player.

- Nothing can be built on a rock. A rock cannot be sold and cannot be chewed.
- Enemies on foot cannot cross a rock. Flyers pass over it.
- The rocks are the same in every session, for every participant, in both versions of the game. They are drawn by hand and not rolled from the seed: the shop is one source of luck already, and the study has no use for a second.

Why rocks, and not walls that are already standing. A wall belongs to the player. It can be sold for budget, and it pays when it is chewed through, so a board that starts with walls starts with money lying on it. A rock is only ground.

What the rocks must not do is decide the maze. The study measures how a player's building changes from wave to wave. If the rocks leave one sensible line, every participant builds that line in both versions and there is nothing left to measure. So:

1. Few: about a tenth of the board. (The board in this file's pictures has 167 rock tiles, which is 11.6 %.)
2. In separate groups, with more gaps between them than a player can afford to close. Which gaps to close is then the player's decision, and the waves give a reason to change it.
3. None close to the spawn or the core.
4. Always a way through on foot. The game checks this when it loads a map, and refuses a map whose rocks cut the spawn off (BUILT).

The rules for rocks are BUILT already: a map can mark tiles as blocked, nothing can be placed on them, the path rule goes round them and flyers ignore them. No map uses them yet. The layout in the pictures is an example only. I draw a first real one in WP-C4, by the four rules above, you change it by playing, and it is checked in WP-C5.

Rocks take the place of the "seeded maze" that the schedule named for the second semester (question 11).

3.3 Size

OPEN (question 19). 38 × 38 is large: a tile is small on the screen, one wall tile changes little, and the walk alone takes over a minute a wave. A smaller board would make every wall matter more and every wave shorter, which also bears on how long a session is (question 1). Worth trying when the numbers are balanced (WP-C5), not before.

3.4 Spawn points

PLANNED **(WP9):** more than one spawn point, because one of the director's four wave types sends groups from different sides. Where the extra spawns go is OPEN (question 10).

4. What you build

4.1 Walls

PLANNED (your decisions, 2026-10-12; in the work plan under WP-C4): **walls are no longer tetromino pieces.** They are single tiles:

- **A wall is one tile.** With the Wall tool, a click builds one tile; holding the button and dragging draws a line of them. Each tile is paid for as it appears, and the line stops when the budget runs out.
- **Every tile costs the same:** 1 budget (a placeholder, like every number here).
- A wall goes on any free tile: not on a rock, a tower, another wall, the spawn or the core. The tile under the mouse shows whether the spot is legal before you click. That preview and the click use one rule, so they can never disagree.
- **A wall is expensive ground, not an obstacle.** To an enemy, stepping onto a wall tile counts like walking 15 tiles. If the way round is longer than that, the enemy goes through: it stands on the tile and chews it (one enemy needs 6 seconds), the tile disappears, and every enemy's route is worked out again. You are paid 1 budget for a tile that is chewed through. With a tile priced at 1 that is the whole price back, which makes the balance question about it sharper (question 9).
- **Thickness is the first thing to watch.** A wall one tile thick costs a walker 15 tiles' worth to cross, and a sapper only 3. Two tiles thick costs 30 and 6.
- **Length is the second.** Walkers go round a wall only while going round is cheaper than chewing. Stretch the detour too far and they chew through the thinnest spot instead. (This happened while the picture in §5.3 was being made: with one baffle three tiles longer, the walkers went through the thin wall marked 1.)
- **Selling.** Between waves, with the Sell tool, a click sells a wall tile for half its price, and a drag sells every wall tile it passes over.
- There is no bag, no queue of next pieces, no rotation and no hold slot.

What the change does:

- **Less to learn.** Two controls and a row of the screen go away. Building is "draw where you want a wall".
- **For the study, one source of luck fewer.** Which piece came next nudged where a player built, in both versions. The measure of rebuilding counts built tiles by region of the board (`ARCHITECTURE.md`

§5.9), so its definition does not change.

- **Smaller answers become possible.** A player can now meet a wave by adding two tiles, where a whole piece was the least before. The measure will see a small patch as a small change. Whether it still tells sessions apart is something to look at in the pilot, before the measure is frozen.
- **What is lost:** the small puzzle of fitting the piece you were dealt.
- **In the code** the rules need no change: a wall "piece" may already have any shape, one tile included. What changes is the list of shapes, the input, the screen, and every recorded session (they have to be recorded again).

BUILT **today**, until WP-C4, is the tetromino version:

- Seven shapes of four tiles each (I, O, T, S, Z, J, L) from a shuffled bag. The current piece and the next three are shown. R rotates the piece; Shift swaps it with a hold slot.
- A piece costs 3 to 7 budget (I 4, O 7, T 5, S 3, Z 3, J 3, L 3) and goes anywhere its four tiles are free.
- A whole piece is sold at once for half its cost, or in proportion less if part of it has been chewed away.

4.2 Towers

BUILT, with a placeholder roster.

- A tower stands on one tile. It is expensive ground too (it counts like 20 tiles of walking), and an enemy that chews through it destroys it. That pays nothing.
- **The shop.** At the start of the game, and each time a wave ends, the shop shows four towers picked at random. Each offer can be bought once. Offers you did not buy stay through the next wave's fight and are replaced when that wave ends. The shop is open during a wave.
- **The shop is the same for everyone on the same seed.** What it offers depends on the seed and the wave number and on nothing else: not on what you built, not on what you bought, not on which version of the game is running.
- **How a tower fights.** Every tower fires on its own. It picks the enemy in range that is nearest to the core (measured along that enemy's own way there), hits it at once, and waits for its next shot. A splash tower also hits everything close to its target. A slowing tower makes its target walk slower for a moment.
- Between waves you can **sell** a tower for half of what you paid.

The three towers in the game today. The numbers are placeholders and nothing is balanced:

Tower	Price	Damage type	What it does	Range	Hits flyers
Archer	10	physical	4 damage every 0.5 s to one enemy	3.5 tiles	yes
Cannon	18	fire	3 damage every 1.2 s to its target and everything within 0.75 tile	2.5 tiles	no
Frost	12	frost	1 damage every 0.6 s; the target moves at half speed for 1.5 s	2.5 tiles	no

PROPOSED **a roster of five for the study** (question 2). With three tower types, a shop of four almost always shows the same tower twice, so there is little to choose. Five types make a roll a real choice, and all five use only rules the game already has:

Tower	Damage type	Role	Hits flyers	Strong against	Weak against
Archer	physical	quick single shots, medium range	yes	flyers, stragglers	a dense swarm
Ballista	physical	one heavy shot, long range, slow to reload	yes	tough enemies, diggers stuck at a wall	a dense swarm
Cannon	fire	splash	no	a swarm on a narrow route	flyers, enemies spread out
Brazier	fire	very short range, fires very fast	no	a route that runs right past it	anything that goes round it or over it
Frost	frost	slows, little damage	no	makes every tower near it better	being alone

The work plan asks for four to six towers covering single target, splash, slow and anti-air across three damage types (WP-C5); this is one way to meet it. Prices and numbers would be set in WP-C5 and tuned in the balance pass.

4.3 How the two work together

A tower only earns its price while enemies are in its range. A wall puts no damage on anyone, but a route bent past the same towers twice doubles what those towers do. So the strong builds are mazes with towers at the bends, and every such build has a weak point somewhere: a wall one tile thick, a single spot that does all the killing, nothing that can shoot upward. Those weak points are what the adaptive version goes looking for.

5. Enemies

5.1 What an enemy is

BUILT An enemy has hit points, a walking speed, a digging speed, and one resistance per damage type (a resistance of 0.5 halves the damage of that type). It ends in one of two ways: **killed** (you are paid 0.2 budget) or **through** to the core (the core loses 1).

5.2 Three ways of moving

BUILT (WP-C2). This is what the director's wave types are made of.

Class	How it gets to the core	Digs	Which towers can hit it
Walker (ground)	Takes the cheapest way, counting a wall as 15 tiles of walking. Usually goes round.	Only when going round costs more.	All
Sapper	The same, but a wall counts as 3 tiles. Goes through walls that walkers go round.	Often, and its group digs faster.	All
Flyer	A straight line to the core, over everything.	Never.	Only towers that hit flyers

Today a sapper is tinted orange and a flyer is blue and drawn above the walls. Those looks are placeholders (WP-C4).

5.3 Where enemies walk FIGURE 1

BUILT (the rule; the game does not draw the lines for sappers and flyers yet). The pink and the orange route in the picture were not drawn by hand: they are what the game's own path rule gives for that board.

Nobody assigns a path to an enemy. There is one rule, and every enemy follows it on its own: *at every step, go to the neighbouring tile from which the rest of the way to the core is cheapest for my kind.* The way is priced in tiles of walking, and a built tile is priced as extra walking: 15 tiles for a walker, 3 for a sapper. A flyer does not use the rule at all. It flies a straight line.

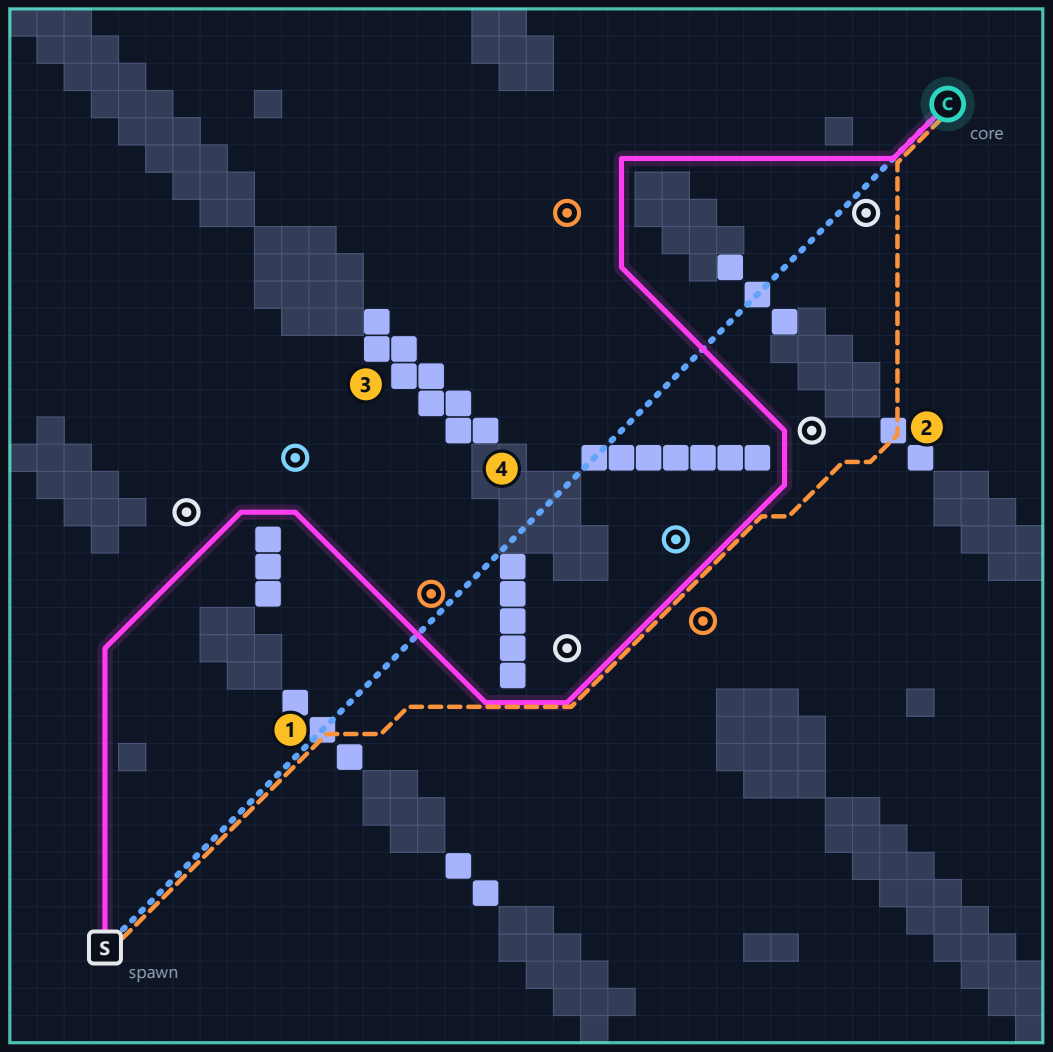
What follows from that:

- **Same kind, same entrance, same route.** All the walkers of a wave come out of the same spawn and walk the same line, one behind the other. They do not spread out, and they do not pick different gaps.
- **Different kinds, different routes, on the same board.** On the board of the picture:

Kind	Its way	Length	Towers it passes
Walker	round every wall	about 71 tiles	all 9
Sapper	through the two walls that are one tile thick	about 50 tiles	5
Flyer	a straight line over everything	44 tiles	1 (of the 4 that can hit flyers)

  So one maze is three different problems, depending on what arrives. That is what the director works with.
- **The director chooses the kind, never the path.** It decides what a wave is made of and, once there is more than one spawn (WP9), from which side it comes. After that each enemy finds its own way by the rule. The director cannot send an enemy to a particular tile, and no enemy is given a path of its own. Wherever an enemy goes, you could have read it off your own board.
- **A route can change in the middle of a wave.** When a wall or a tower is chewed through, or when you build, every route is worked out again and every enemy carries on from the tile it stands on.
- **A route is never closed.** Wall off every gap, and the wave chews through the cheapest spot.

FIGURE 1 **5.3 Where enemies walk**



Where enemies walk

One board, three kinds of enemy, three routes. Nobody steers an enemy: each one takes the cheapest way to the core for its own kind.

Walkers go round

To a walker, a wall tile counts like 15 tiles of walking. Going round is cheaper here, so they take the long way: about 71 tiles, past all nine towers. They chew nothing.

Sappers go through

To a sapper, a wall tile counts like 3 tiles. It cuts through the two walls that are one tile thick (1 and 2), leaves the thick wall (3) alone, and is there after about 50 tiles, past five towers.

Flyers go straight

A straight line over rocks (4), walls and towers: 44 tiles. Nothing you build bends it. Only towers that can hit flyers shoot at them; here one of those is in reach.

Same kind, same entrance, same route.

Enemies of one kind walk in single file along one line. The line is drawn on the board before the wave, so you can build against it.

The director chooses the kind, never the path.

It decides what comes (and later, from which side). After that every enemy finds its own way by the rule above. It cannot aim one at a tile.

A route can change in mid-wave.

When a wall is chewed through, or you build, every enemy on the board works out its way again from where it stands.

Dark blocks: rocks, part of the map. Light tiles: walls the player drew. Rings: towers. The pink and orange routes are what the game's own path rule gives for this board.

Where the three kinds of enemy walk on one board

What the screen shows. BUILT one line, the walkers' (pink). PLANNED (your decision, 2026-10-12; WP-C4): in a build phase, one line for every kind of enemy you have met so far in this session; during a wave, the lines of the kinds in that wave. The lines are worked out again each time you build. Drawing all three from the first wave would be simpler, but in the static version it would warn of enemies that never come. With "the kinds you have met", the rule is the same in both versions and the static version keeps the one line it has today. Your supervisor should still see this before the pilot, because it is one of the things the screens of the two versions differ by.

5.4 Enemy types

PLANNED **(WP-C5):** three or four named types, each a fixed set of numbers on top of a movement class. PROPOSED names and roles (question 3):

Type	Class	Compared with the plain walker
Walker	ground	the baseline
Swarmling	ground	many more of them, fewer hit points each, faster
Sapper	sapper	fewer, tougher, slower, digs four times as fast
Flyer	flying	fewer, half the hit points, faster

The sapper and flyer numbers above are the placeholders used by the development planner today. Which waves use resistances is OPEN (question 4).

6. Waves

6.1 The static version

BUILT This is the baseline condition of the study. Every wave is one stream of walkers from the one spawn, and every wave is a little worse than the one before:

Wave	Enemies	Hit points each	Speed (tiles per second)	Gap between enemies
1	100	10	0.63	0.20 s
5	120	18	0.70	0.15 s
10	145	28	0.79	0.15 s
15	170	38	0.89	0.15 s
20	195	48	0.98	0.15 s
25	220	58	1.08	0.15 s

(+5 enemies, +2 hit points and a little more speed each wave.) These numbers are placeholders until the balance pass.

6.2 The adaptive version

PLANNED (WP8 to WP11). Nothing of the director is built yet. What exists is a development planner that sends walkers, sappers, flyers and a mix in a fixed cycle, so that the three classes can be played.

What the director looks at. Once per wave, at the moment the build phase ends, it measures five things about your build:

Measure	In plain words
maze length	how much longer the route is than on an empty board
breach vulnerability	how much of the route depends on walls or towers that could be chewed through
chokepoint reliance	whether your damage is done in one spot or spread along the route
tower concentration	whether your towers stand together or apart
damage type mix	whether your damage is one type or several

Their exact formulas are OPEN (question 5).

What it can send. One of four wave types. The third column is what each is *meant* to press on. Which builds each one really moves a player to change is not a rule of the game: it is what the study measures.

Wave type	What arrives	Meant to press on
Breach the thin wall	sappers	a long maze held by thin walls
Swarm the chokepoint	many weak, fast enemies packed together	all the damage being done in one spot
Split groups	groups from more than one spawn	towers standing together in one place
Flying bypass	flyers	relying on the maze, with nothing that shoots upward

How many enemies of which type each wave type contains is OPEN (question 4).

What keeps it fair. These are fixed rules of the project (`CLAUDE.md` §4):

- **Same strength.** Every wave type costs exactly what the static wave of that number costs, on one shared price list for enemies. The director chooses the shape of a wave, never its size.
- **Nothing it cannot be answered with.** Flyers are not sent while you own no tower that can hit them and the shop is not offering one.
- **No repeats in disguise.** A wave too similar to the previous one is ruled out.
- **Hands off your side.** The director never changes the shop, a price, a tower or your budget. (Today one thing leaks through: you are paid per kill, so a wave of fewer, tougher enemies pays less. That has to be closed before the wave types are priced, question 6.)

How it chooses. It starts from written rules: each wave type gets a score for the build in front of it. On top of that it learns, during your session, which wave types actually made *you* rebuild, and nudges close calls that way. The nudge is small by construction: it can re-rank wave types that scored about the same, never overrule a clear one. The player sees none of these numbers.

What it is rewarded for is not beating you. It scores a wave highly when you changed your build afterwards, when your core was still standing, when the wave got close to the core, and when the wave was different from the last one. A wave that ends your session scores badly, because a player who is out builds nothing more.

The banner. As each wave starts, one line of text says what is coming and what in your build it is aimed at. The lines are written by hand, 20 of them (four wave types × five situations), and never generated. Sample wording, PROPOSED:

Wave type	Sample line
Breach the thin wall	"Sappers. They dig where your wall is one tile thick."
Swarm the chokepoint	"A swarm. Too many for one kill zone."
Split groups	"Two groups, two directions. Your towers are all in one place."
Flying bypass	"Flyers. Your maze does not reach the sky."

When the banner appears. The director has to see your finished build, so it chooses at the moment the build phase ends. The line therefore appears as the wave starts, not while you are building. You answer it during the wave (walls and towers can be placed in mid-wave) and in the next build phase. Whether that is enough warning to feel fair should be playtested (question 7). The route lines of §5.3 help here: from the moment the wave starts, the banner names a line that is drawn on the board, not something you have to imagine.

6.3 What is the same in both versions

The board and its rocks, the shop's offers (on the same seed), every price and reward, the strength of every wave, and the screen. Only the shape of the waves differs, and with it the text of the banner (question 7) and which route lines there are to draw (§5.3).

7. Money

BUILT One budget pays for walls and towers. All numbers are placeholders.

	Amount
At the start	60
After every wave	+20
An enemy killed	+0.2 (wave 1 killed to the last enemy: +20)
A wall tile chewed through	+1
A tower chewed through	nothing
A wall	today −3 to −7 for a piece of four tiles; planned −1 a tile
A tower	−10 to −18
Selling a tower	half its price back
Selling a wall	half its cost back (today a whole piece at once, in proportion to the tiles still standing; planned tile by tile)
Rerolling the shop	not in the game (the rule exists and is switched off)

Sixty at the start is about four towers, or three towers and some twenty wall tiles at that price.

8. Losing, winning, and a session

BUILT The core has 10 hit points and loses one for every enemy that reaches it. At zero the game is over and the screen shows "You reached Wave N" and a Restart button. There is no way to win: the waves go on until the core falls.

OPEN **(question 1): how long is a session?** This is the most important open question in this file. The study compares how a player builds over the waves they *completed* in each of their two sessions, cut to the shorter one. A participant who falls in wave 4 of one session leaves three waves to compare. So the study needs a stated length (the project's notes assume about 25 waves) and a game balanced so that an ordinary player gets most of the way there.

PLANNED **(WP14): a study session.** The participant types a code on a start screen. The code decides their two sessions: one static, one adaptive, on two different seeds, in an order that differs between four groups of participants. Nobody plays the same seed twice, so nobody can remember the shop.

OPEN **(question 8):** how a participant learns the game before their first session.

9. What the screen looks like

9.1 Today FIGURE 2

BUILT This is the earlier design, before the changes of 2026-10-12: a tilted 3-D view of the board that can be turned, moved and zoomed, and walls that are tetromino pieces. Top left: the core's health. Top centre: the wave and the countdown. Bottom: the budget, the hold slot, the current piece and the next three, and the Start, speed and Pause buttons. Left: the list of controls. The pink line is the route the walkers will take. Walls are glowing cubes in the colour of their shape; towers are cubes tinted by damage type; enemies are small cubes with a health bar. The text box at the top right is the temporary shop.

9.2 The target: a build phase FIGURE 3

A mockup, on the board of the picture in §5.3. What in it is decided and what is not:

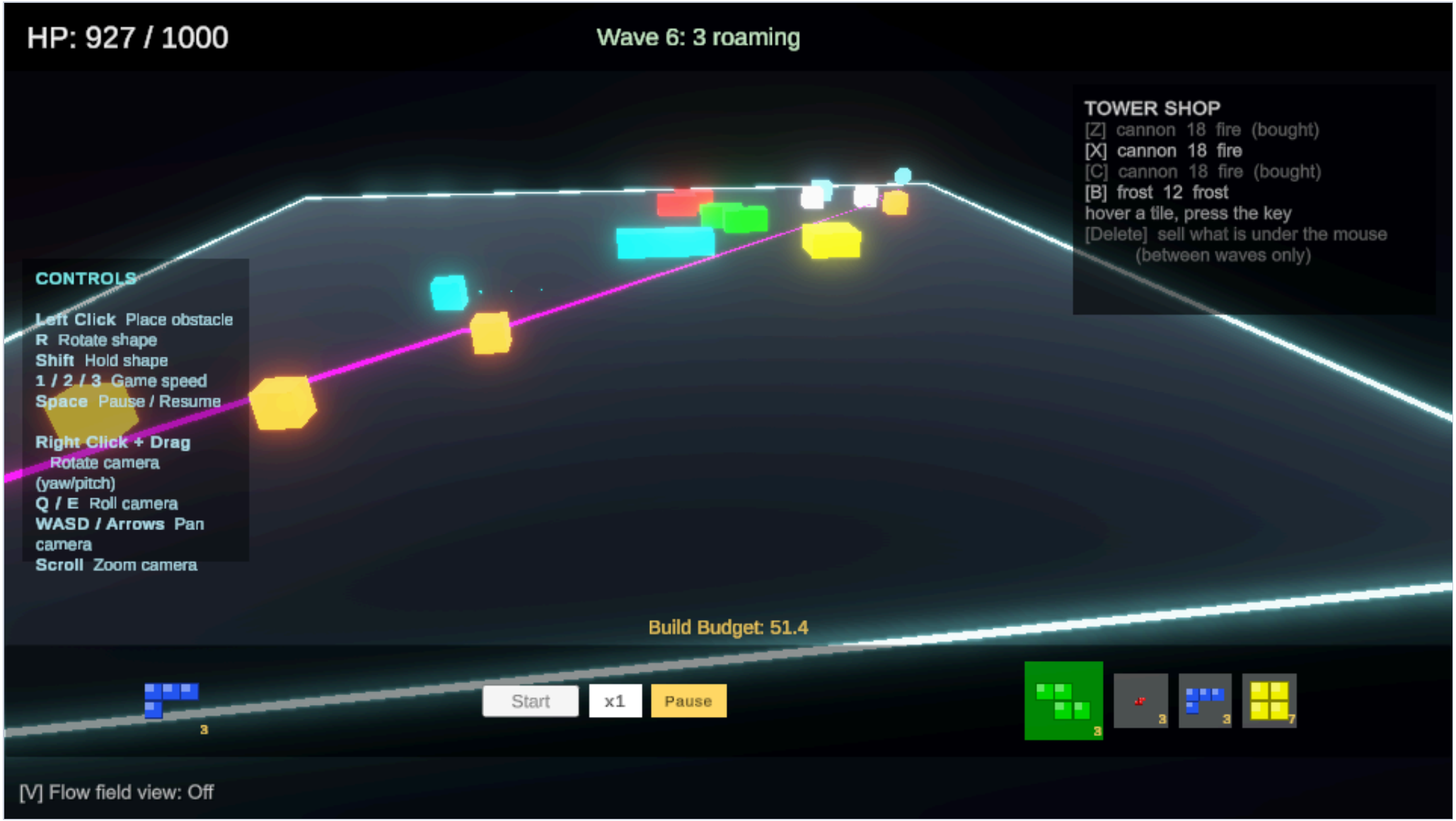
On the screen	Status
The board seen from straight above	PLANNED (your decision)
Walls as single tiles, drawn with the Wall tool	PLANNED (your decision, §4.1)
Rocks	PLANNED (your decision, §3.2); this layout is only an example
A route line for each kind of enemy you have met, with a key	PLANNED (your decision, §5.3)
The shop as four cards; a bought tower follows the mouse with its range ring	PLANNED (WP-C4)
Sell as a tool that shows what would be sold, and the refund	PLANNED (WP-C4)
Where things are on the screen	PROPOSED

- **The shop.** Click a card, and the tower follows the mouse with its **range ring**; click a tile to place it. Esc puts it down.
- **Wall and Sell are tools**, and one is on at a time. With Sell on, pointing at a tower or a wall tile shows the refund, and a click sells it.

9.3 The target: a wave, in the adaptive version FIGURE 4

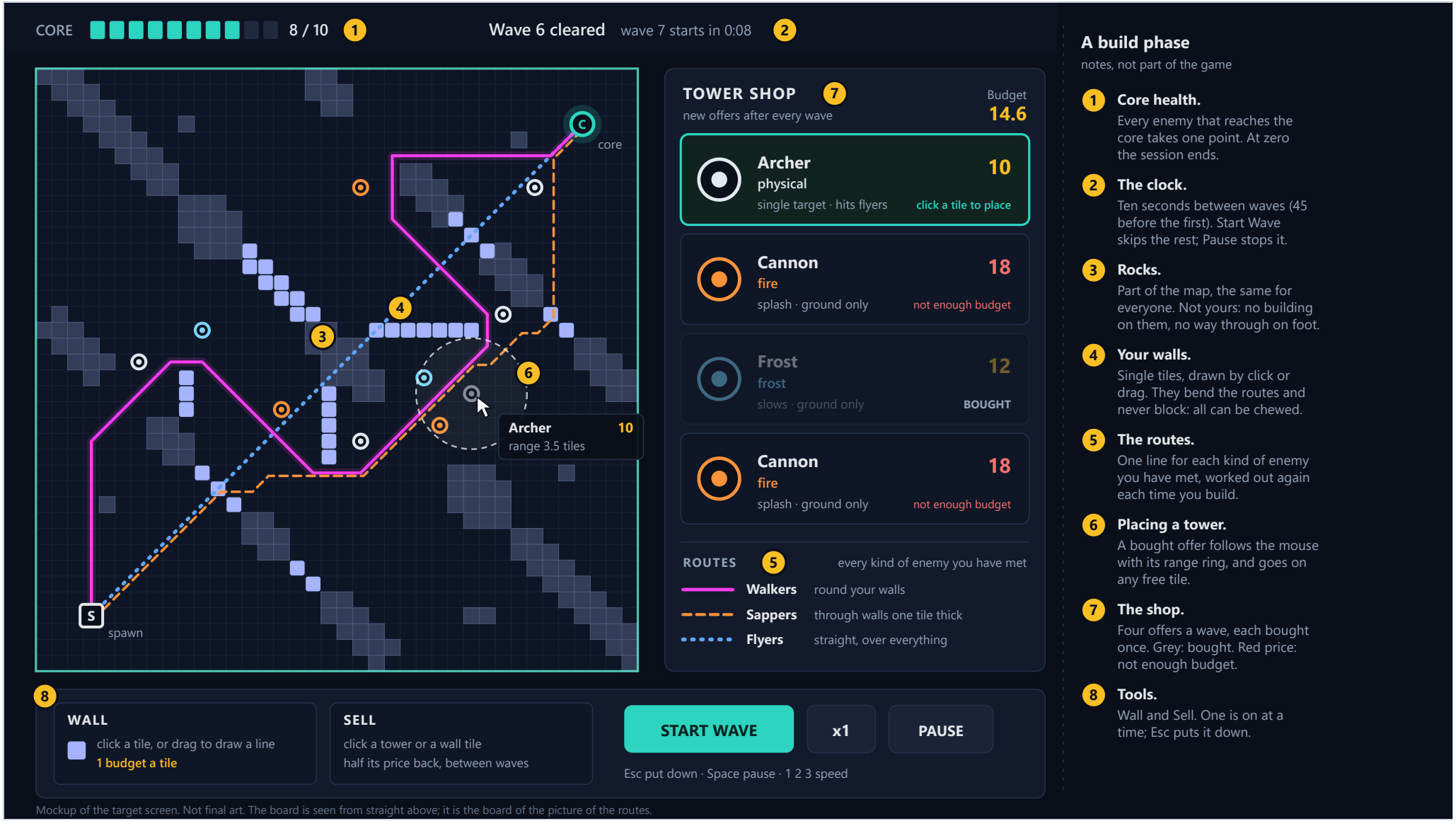
- The **banner** (PLANNED , WP14).
- Only the route of what is coming is drawn: here the sappers' (PLANNED , §5.3). One of the two thin walls is already chewed through, and the first sappers are at the second.
- Towers that turn towards their target and fire a visible shot (PLANNED , WP-C4). The shot is only drawn; the damage is done the instant the tower fires.
- An enemy model per type, in place of today's tint (PLANNED , WP-C4).
- The Sell tool is visibly off until the wave is over (PLANNED , WP-C4).

FIGURE 2 **9.1 Today**



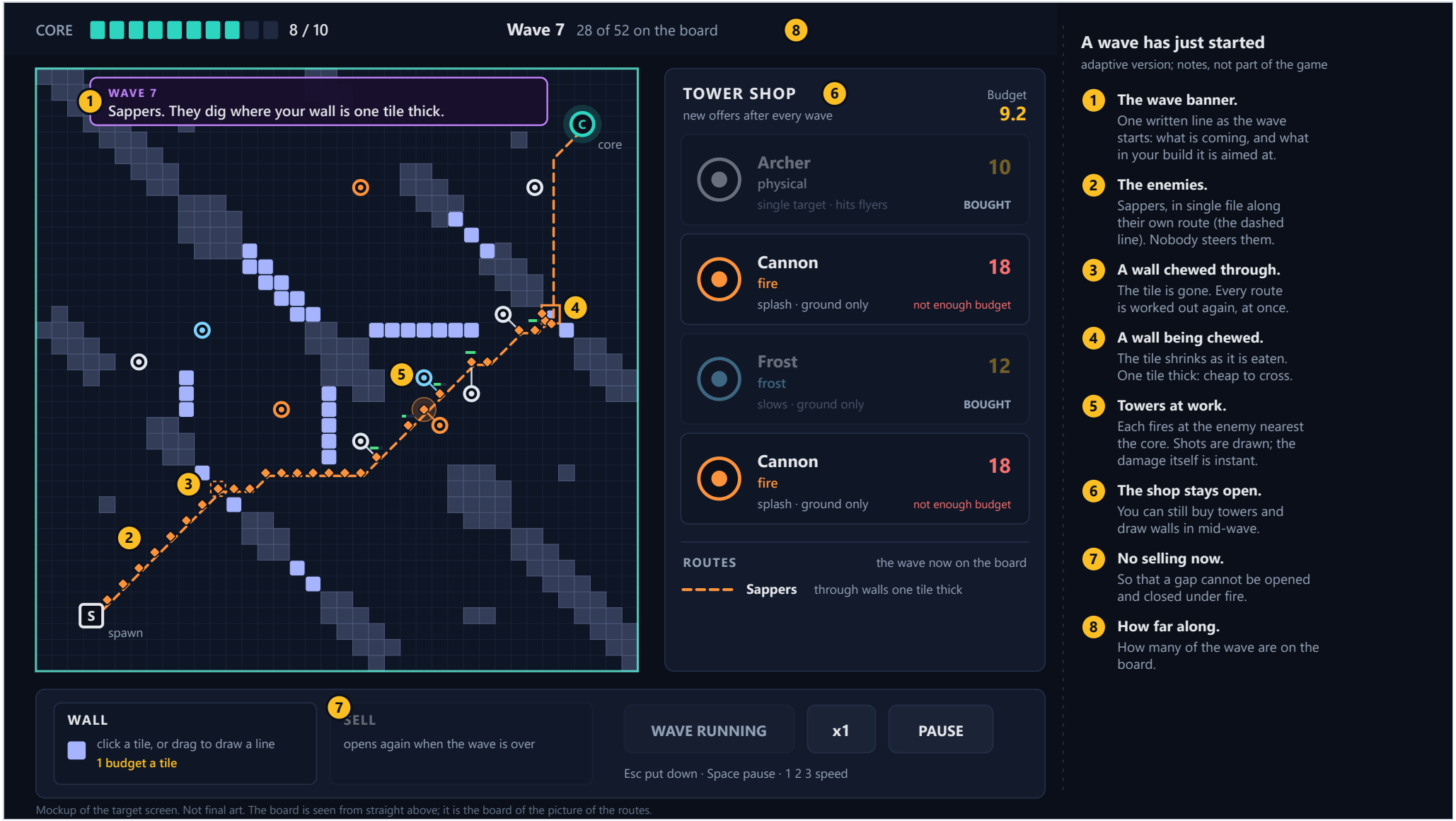
The game on 2026-10-11

FIGURE 3 **9.2 The target: a build phase**



Mockup of a build phase

FIGURE 4 **9.3 The target: a wave, in the adaptive version**



Mockup of a wave that has just started

9.4 Other screens and overlays

Screen	Status
Game over: "You reached Wave N", Restart	BUILT
Flow-field view (key V): arrows, a heat map, or both, showing which way every tile leads	BUILT
Session start: participant code	PLANNED (WP14)
Director overlay (F9): why the director chose what it chose. For development; not in a participant's build	PLANNED (WP14)
Preview of how the routes would change, before a wall is paid for	PLANNED (second-semester UX pass)

10. Controls

Action	Today (BUILT)	Target
Build a wall	left click places the current piece	Wall tool: click a tile, or drag to draw a line (PLANNED , your decision)
Rotate the piece; swap it with the hold slot	R; Shift	gone with the pieces
Buy a tower	Z, X, C or B (offers 1 to 4) with the mouse over a tile	click the card, then click a tile (PLANNED , WP-C4)
Sell a tower or a wall	Delete or Backspace with the mouse over it, between waves; a wall goes as a whole piece	Sell tool: click it, or drag over wall tiles (PLANNED , WP-C4)
Put a tool or a tower down	(nothing to put down)	Esc
Start the wave early	Start button	the same
Game speed	1, 2, 3	the same
Pause (building is off while paused)	Space	the same
Turn, move, zoom and roll the camera	right mouse button and drag; W A S D or the arrow keys; mouse wheel; Q, E and C	gone: the board is fixed on the screen (PLANNED , your decision)
Flow-field view	V	the same, for development

For a participant that is seven controls where there were twelve.

Known clash today: C both buys offer 3 and resets the camera's roll. It goes away with the keys.

11. Look and sound

Look. BUILT, as a placeholder: a dark board with a glowing edge, and everything on it a glowing cube. It is cheap, it reads well from above, and it is what the first pictures of the game look like. Rocks should look like ground and not like something built: darker and duller than a wall.

PLANNED **for the study build:** simple distinct shapes, so that a tower type and an enemy type can be told apart without reading (WP-C4), and then a pass over the whole screen in the second semester. Art beyond that is out of scope.

Sound. OPEN (question 13). There is none today and none in the plan.

12. Open questions

Each of these changes what the finished game is. "Suggestion" is this file's proposal, not a decision. Questions 15 to 19 came out of your first read on 2026-10-12. You answered 15 to 18 the same day; they are under the table.

#	Question	Suggestion	Who decides	Needed by
1	**How long is a session, and what ends it?**	A fixed length of 25 waves; the session still ends early if the core falls; balance so that an ordinary player gets past wave 15. Reaching the last wave is the win.	you and your supervisor	the balance check (WP-C5), firmly before the pilot
2	**Which towers?** (spec gap S12)	The five in §4.2.	you	WP-C5
3	**Which enemy types?** (S12)	The four in §5.4.	you	WP-C5
4	**What exactly is in each of the four wave types,** and do they use resistances? (S10)	Take it from the proposal's §6.3, which is not in the repository yet.	you (add the proposal to `Docs/`)	WP9
5	**The exact formulas of the five measures.** (S9)	Take them from the proposal's §6.2. Starting definitions are in `ARCHITECTURE.md` §5.3.	you (add the proposal to `Docs/`)	WP8
6	**Income must not depend on the wave's shape.** (S14)	Pay a kill in proportion to what that enemy costs on the shared price list, so a wave that is wiped out pays the same whatever it was made of.	delegated; I will propose it with WP9	WP9
7	**The banner.** Does the static version show one? Is "as the wave starts" enough warning?	Both versions show a banner. The static one states the wave plainly ("Wave 7. 130 walkers."); the adaptive one also says why. Give every wave a few seconds between the banner and the first enemy, and playtest it.	you and your supervisor (it is part of the study design)	WP14
8	**How does a participant learn the game?**	A short practice run of three waves on a third seed, static, not analysed.	you and your supervisor	the pilot

#	Question	Suggestion	Who decides	Needed by
9	**Balance questions:** does a chewed wall keep paying +1; the refund of a sale; rerolls.	Leave as they are until the balance check has numbers.	delegated	WP-C5, then the second-semester balance pass
10	**Where do the extra spawn points go?** (S8)	The two corners the spawn and the core do not use, to be checked against the price list (from there the walk to the core is shorter).	delegated	WP9
11	**Is the "seeded maze" still wanted?**	No. The rocks of §3.2 do its job (the player does not start from an empty board), and they are fixed, so nothing new is random.	you	any time (the rocks are decided)
12	**A name for the game,** and the company and product name in the build settings (still `DefaultCompany/InternProj`). They decide where a build writes its data.	—	you	the first build that records data
13	**Sound.**	A handful of effects (place, shot, enemy through, wave start) in the UX pass, only if there is time. Not needed for the study.	you	the second-semester UX pass
14	**May a participant pause and change speed?** They can today.	Yes to both: pause is for thinking, and building is off while paused.	you and your supervisor	the pilot
19	**How big is the board?**	Try a smaller one when the numbers are balanced.	you	WP-C5

Answered

#	Question	Your answer	When
15	The walls, in detail	Yes, as in §4.1: single tiles drawn by click and drag, one price a tile, sold tile by tile.	2026-10-12
16	Rocks: yes or no, and where?	Yes, by the four rules of §3.2. A first layout is drawn in WP-C4 and you change it by playing.	2026-10-12
17	Is the wave watched from above too?	Yes: one view for the whole game.	2026-10-12
18	Which route lines are drawn?	One for each kind of enemy met so far in the session (§5.3). Still to be shown to your supervisor before the pilot, because the screens of the two versions differ by it.	2026-10-12

13. From today to the finished game

What each coming package changes for the person playing.

Package	What the player will notice
WP-C4 towers and shop on screen (next)	The board from above, with rocks on it; walls drawn tile by tile; a route line for every kind of enemy you have met. The shop as cards, a range ring before placing, a Sell tool with a preview, towers that turn and fire, a model per enemy type.
WP-C5 roster and balance check	The real towers and enemy types; numbers that are not absurd.
WP7 the learner	Nothing. It runs without the game.
WP8 build profile	Nothing.
WP9 wave types	The four wave types exist; more than one spawn.
WP10, WP11 the director	Waves that answer the build.
WP14 director on screen, session start	The banner; the participant-code screen.
WP12, WP13 harness and gates	Nothing.
Second semester: UX pass, balance pass	Route previews, a clearer screen, a game that is fair to an ordinary player.
Build freeze	Nothing changes after this, except crash fixes.

What the changes of 2026-10-12 cost. They belong in WP-C4, which rebuilds the building input and the screen anyway. Some of its work falls away (no piece queue, rotation or hold slot to carry over; a sale is one tile) and some is added (a fixed camera, drawing by drag, a map with rocks, two more route lines, and recording the pinned sessions again). My estimate is two to three days more than WP-C4 had; it is an estimate, not a measurement. The plan has no slack left (`WORKPLAN.md`), so that is time the later packages do not have, and the cut order in `CLAUDE.md` §9 says where it would come from.

14. What this game is not

- **Not a game that gets easier or harder to suit you.** Difficulty is held equal on purpose; only the shape of the waves changes.
- **Not a game where walls block.** That was the original design and was removed; it is what makes "a path always exists" true.
- **Not a puzzle of fitting shapes.** That was the tetromino version of the walls (§4.1).
- **No generated text or behaviour.** Every banner line is written by hand, and every run can be replayed exactly.
- **Not in the plan:** tower upgrades, more than one map, anything carried over from one session to the next, more than one player.